LAB EXPERIMENT RECORD

Set 30: Mobile App Usage

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Q1. Histogram and Density Plot for Screen Time

Aim: To visualize and interpret the distribution of mobile screen time.

R Code:

R

```
# Create dataset

data <- data.frame(

  User_ID = c("U01", "U02", "U03", "U04", "U05", "U06"),

  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),

  Age = c(20, 22, 19, 21, 23, 20),

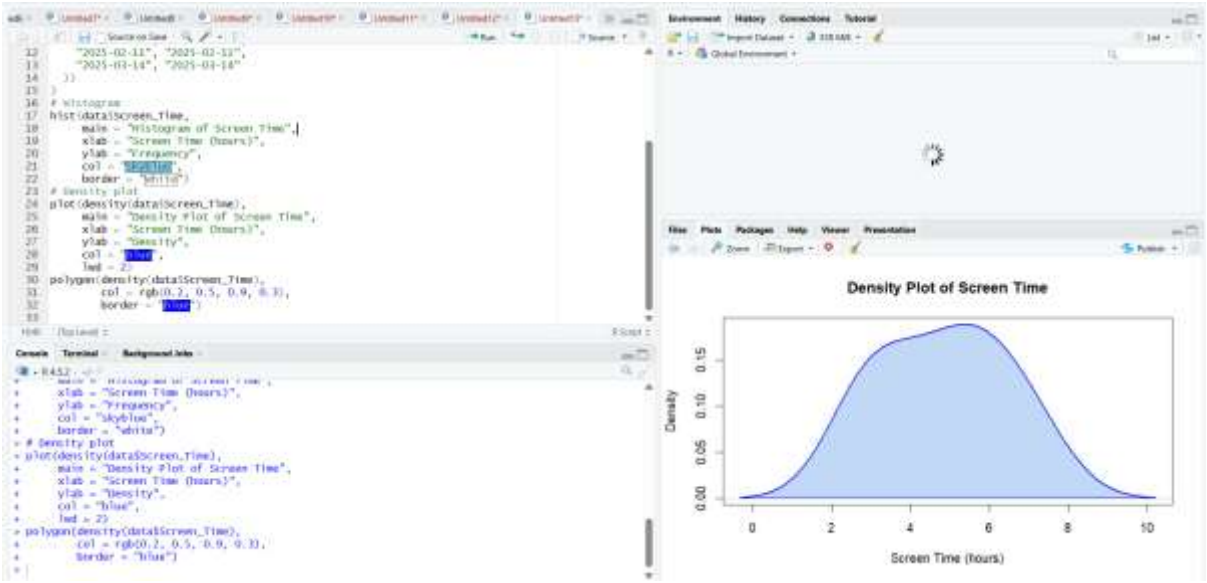
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),

  App_Usage_Count = c(18, 25, 12, 30, 10, 22),

  Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),

  Satisfaction = c(3, 5, 3, 5, 2, 4),
```

```
  Usage_Date = as.Date(c(
    "2025-01-08", "2025-01-08",
    "2025-02-11", "2025-02-11",
    "2025-03-14", "2025-03-14"
  ))
)
# Histogram
hist(data$Screen_Time,
     main = "Histogram of Screen Time",
     xlab = "Screen Time (hours)",
     ylab = "Frequency",
     col = "skyblue",
     border = "white")
# Density plot
plot(density(data$Screen_Time),
     main = "Density Plot of Screen Time",
     xlab = "Screen Time (hours)",
     ylab = "Density",
     col = "blue",
     lwd = 2)
polygon(density(data$Screen_Time),
        col = rgb(0.2, 0.5, 0.9, 0.3),
        border = "blue")
```



Observation: Screen time ranges from 2.8 to 7.1 hours. The users show variation in daily screen time, but six observations are too few to establish a reliable overall distribution.

Result: The histogram and density plot visualize the distribution of screen time.

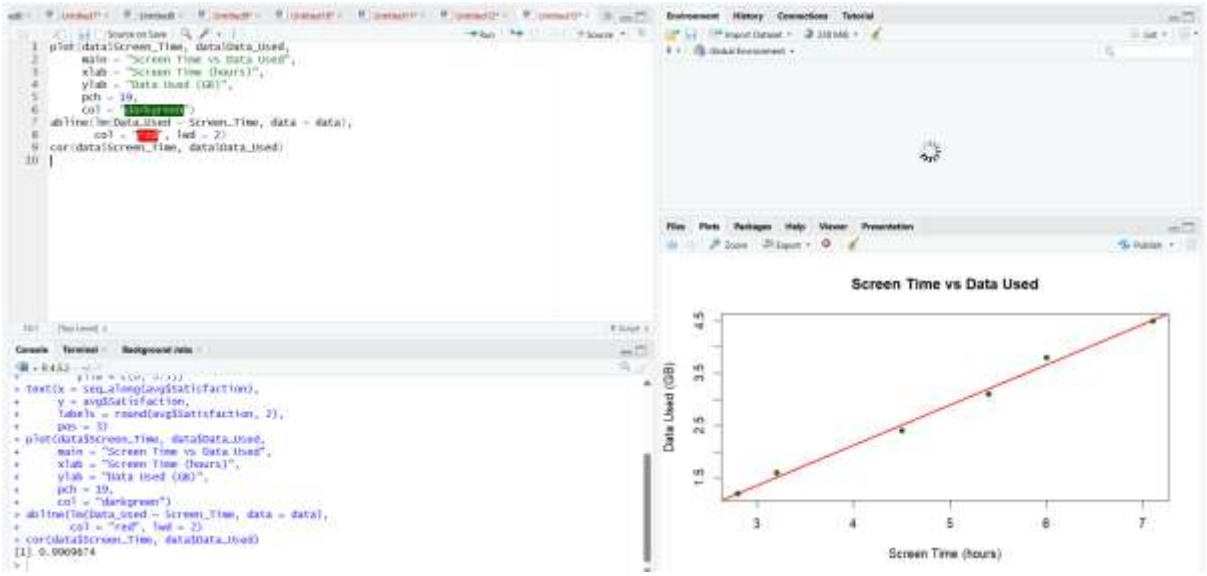
Q2. Scatter Plot of Data Used and Screen Time

Aim: To examine the relationship between screen time and mobile data usage.

R Code:

R

```
plot(data$Screen_Time, data$Data_Used,
     main = "Screen Time vs Data Used",
     xlab = "Screen Time (hours)",
     ylab = "Data Used (GB)",
     pch = 19,
     col = "darkgreen")
abline(lm(Data_Used ~ Screen_Time, data = data),
       col = "red", lwd = 2)
cor(data$Screen_Time, data$Data_Used)
```



Observation: Screen time and data usage have a strong positive correlation, approximately **0.99**. Users with higher screen time generally consume more mobile data in this dataset.

Result: The scatter plot and correlation coefficient show a strong positive relationship between screen time and data usage.

Q3. Average Satisfaction by Gender

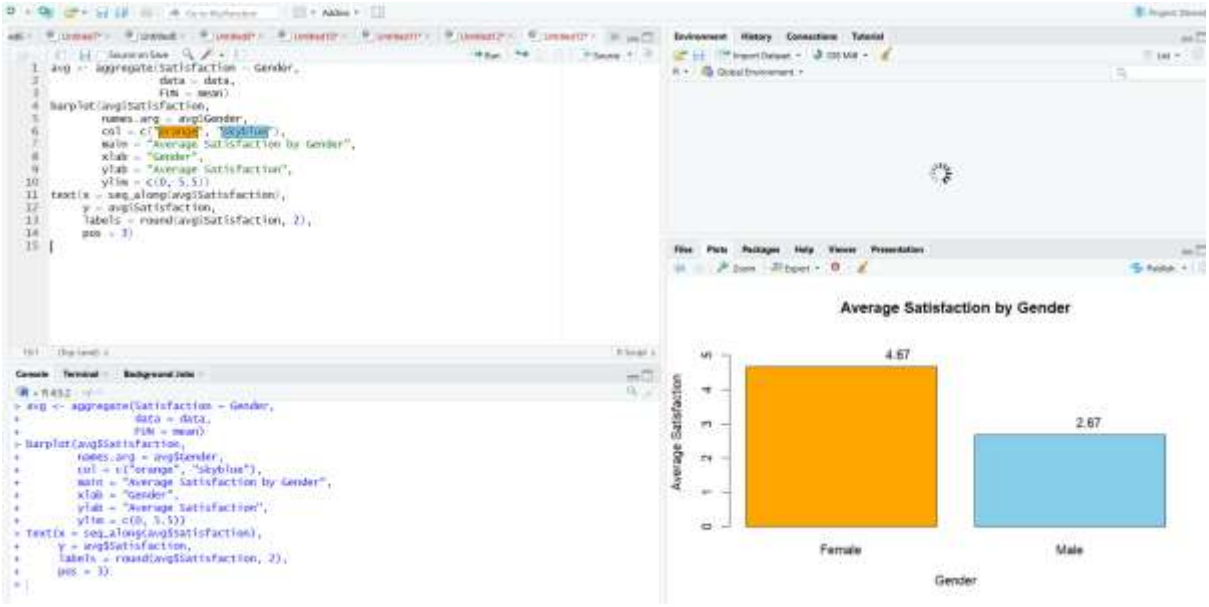
Aim: To calculate average satisfaction scores for each gender and display them in a bar chart.

R Code:

R

```
avg <- aggregate(Satisfaction ~ Gender,
                 data = data,
                 FUN = mean)
barplot(avg$Satisfaction,
        names.arg = avg$Gender,
        col = c("orange", "skyblue"),
        main = "Average Satisfaction by Gender",
        xlab = "Gender",
```

```
    ylab = "Average Satisfaction",
    ylim = c(0, 5.5))
text(x = seq_along(avg$Satisfaction),
   y = avg$Satisfaction,
   labels = round(avg$Satisfaction, 2),
   pos = 3)
```



Observation: The average satisfaction score for Male users is 2.67, while Female users have an average of 4.67. Female users report higher satisfaction in this small sample.

Result: The bar chart compares average satisfaction scores between genders.

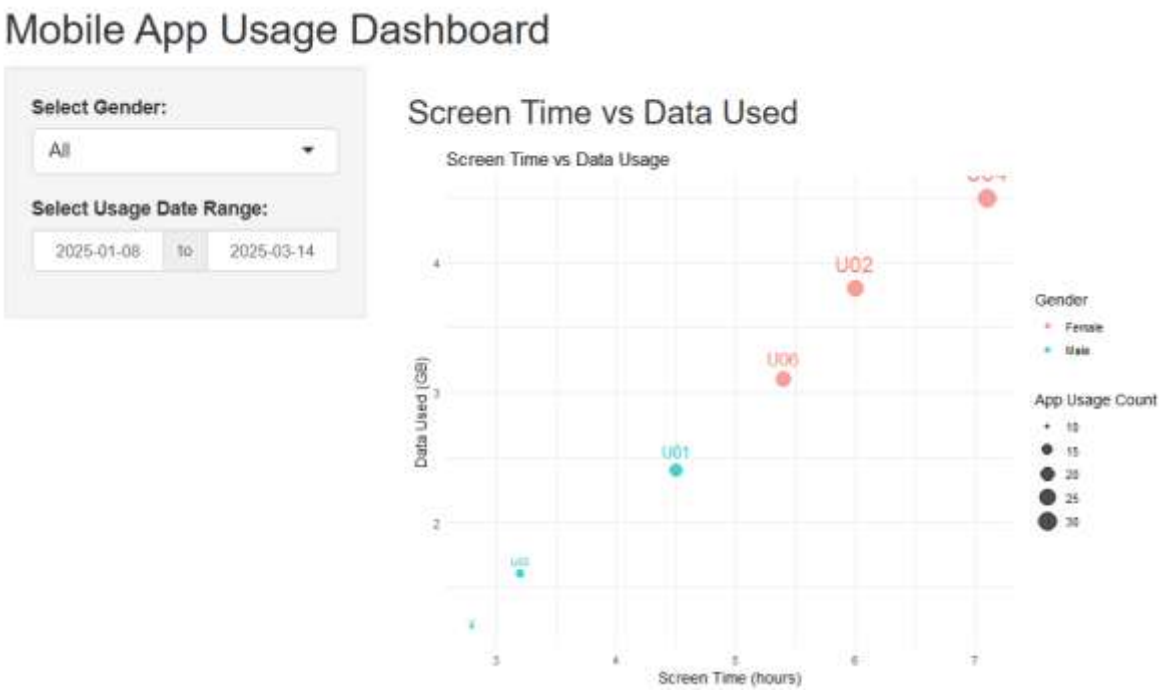
Q4. Tableau Mini Dashboard

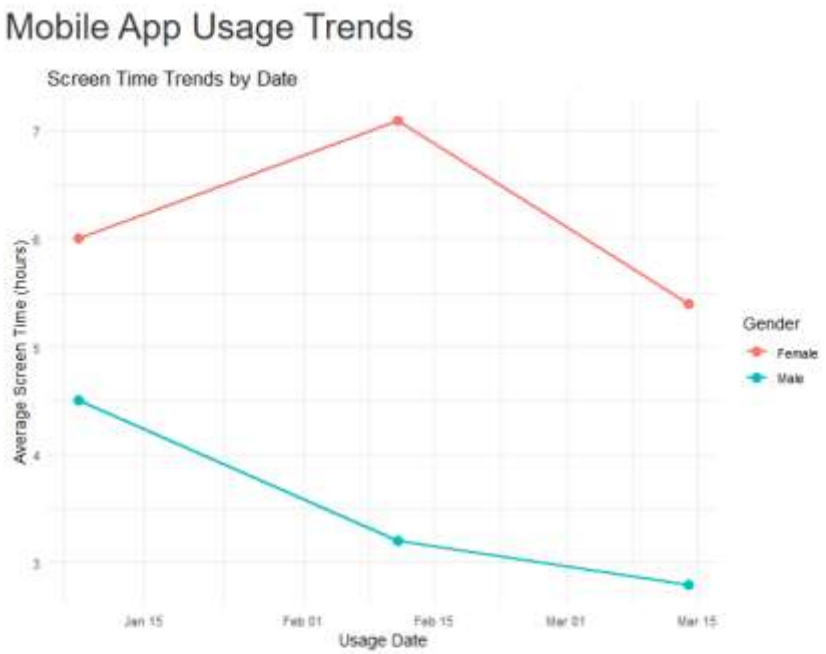
Aim: To create a bubble chart and line chart for mobile app usage, with filters for gender and usage date.

Procedure:

1. Save the dataset as Mobile_App_Usage.csv and import it into Tableau.
2. Create a bubble chart by dragging Screen_Time to Columns and Data_Used to Rows.

3. Drag Gender to Color, App_Usage_Count to Size, and User_ID to Detail.
4. Create a line chart with Usage_Date on Columns and Screen_Time on Rows. Add Gender to Color.
5. Add Gender and Usage_Date as filters and select **Show Filter**.
6. Create a dashboard and place both charts together.





Filtered User Details

User_ID	Gender	Age	Screen_Time	App_Usage_Count	Data_Used	Satisfaction	Usage_Date
U01	Male	20.00	4.50	18.00	2.40	3.00	20096.00
U02	Female	22.00	6.00	25.00	3.80	5.00	20096.00
U03	Male	19.00	3.20	12.00	1.60	3.00	20130.00
U04	Female	21.00	7.10	30.00	4.50	5.00	20130.00
U05	Male	23.00	2.80	10.00	1.20	2.00	20161.00

Analytical Interpretation:

- U04 has the highest screen time at 7.1 hours and data usage at 4.5 GB.
- U05 has the lowest screen time at 2.8 hours and data usage at 1.2 GB.
- The bubble size highlights users with higher app usage counts.
- The line chart compares usage patterns across the three dates.
- Gender and date filters allow users to explore different groups and time periods.

Result: The Tableau dashboard provides an interactive view of screen time, data consumption, app usage, and usage trends.